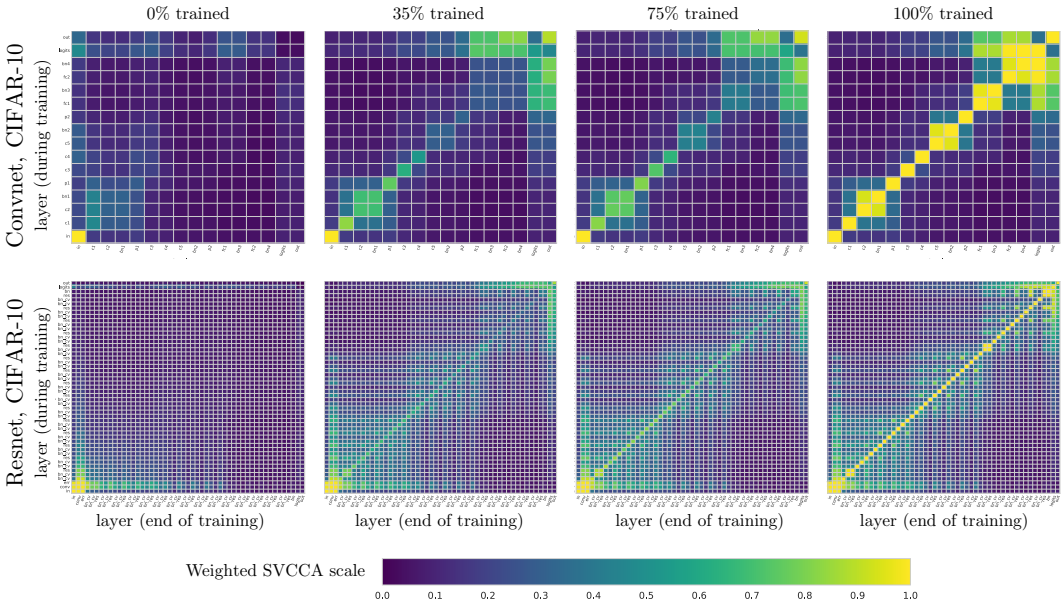

Figure 4: Learning dynamics plots for conv (top) and res (bottom) nets trained on CIFAR-10. Each pane is a matrix of size layers $\times$ layers, with each entry showing the SVCCA similarity $\bar{\rho}$ between the two layers. Note that learning broadly happens 'bottom up' – layers closer to the input seem to solidify into their final representations with the exception of the very top layers. Per layer plots are included in the Appendix. Other patterns are also visible – batch norm layers maintain nearly perfect similarity to the layer preceding them due to scaling invariance (with a slight reduction since batch norm changes the SVD directions which capture 99% of the variance). In the resnet plot, we see a stripe like pattern due to skip connections inducing high similarities to previous layers.

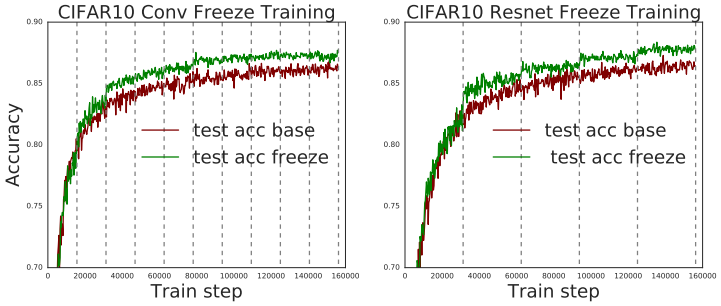

Figure 5: Freeze Training reduces training cost and improves generalization. We apply Freeze Training to a convolutional network on CIFAR-10 and a residual network on CIFAR-10. As shown by the grey dotted lines (which indicate the timestep at which another layer is frozen), both networks have a 'linear' freezing regime: for the convolutional network, we freeze individual layers at evenly spaced timesteps throughout training. For the residual network, we freeze entire residual blocks at each freeze step. The curves were averaged over ten runs.

4.4 Other Applications: Cross Model Comparison and compression

SVCCA similarity can also be used to compare the similarity of representations across different random initializations, and even different architectures. We compare convolutional networks on CIFAR-10 across random initializations (Appendix) and also a convolutional network to a residual network in Figure 7, using the DFT method described in 3.

In Figure 3, we saw that projecting onto the subspace of the top few SVCCA directions resulted in comparable accuracy. This observations motivates an approach to model compression. In particular, letting the output vector of layer l be $\boldsymbol{x}^{(l)} \in \mathbb{R}^{n \times 1}$, and the weights $W^{(l)}$, we replace the usual $W^{(l)}\boldsymbol{x}^{(l)}$ with $(W^{(l)}P_x^T)(P_x\boldsymbol{x}^{(l)})$ where P_x is a $k \times n$ projection matrix, projecting $\boldsymbol{x}$ onto the top SVCCA directions. This bottleneck reduces both parameter count and inference computational cost

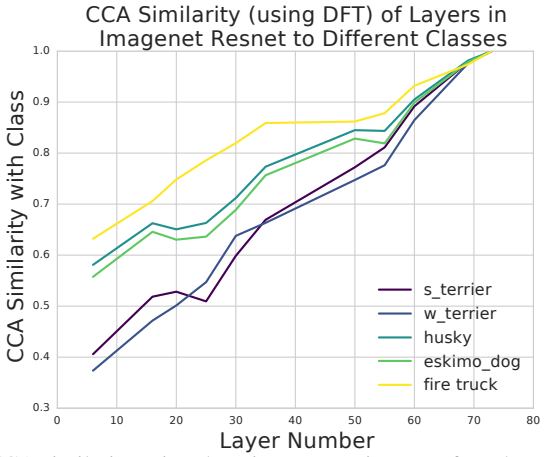

Figure 6: We plot the CCA similarity using the Discrete Fourier Transform between the logits of five classes and layers in the Imagenet Resnet. The classes are firetruck and two pairs of dog breeds (terriers and husky like dogs: husky and eskimo dog) that are chosen to be similar to each other. These semantic properties are captured in CCA similarity, where we see that the line corresponding to firetruck is clearly distinct from the two pairs of dog breeds, and the two lines in each pair are both very close to each other, reflecting the fact that each pair consists of visually similar looking images. Firetruck also appears to be *easier* for the network to learn, with greater sensitivity displayed much sooner.

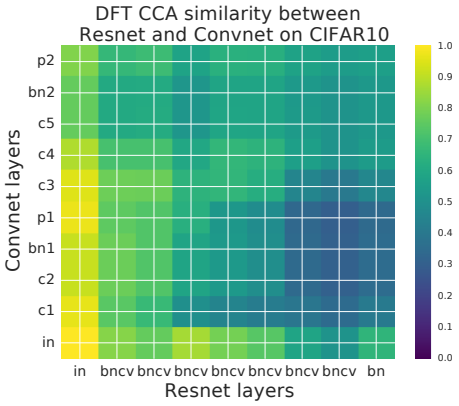

Figure 7: We plot the CCA similarity using the Discrete Fourier Transform between convolutional layers of a Resnet and Convnet trained on CIFAR-10. We find that the lower layrs of both models are noticeably similar to each other, and get progressively less similar as we compare higher layers. Note that the highest layers of the resnet are least similar to the lower layers of the convnet.

for the layer by a factor $\sim \frac{k}{n}$. In Figure App.5 in the Appendix, we show that we can *consecutively* compress top layers with SVCCA by a significant amount (in one case reducing each layer to 0.35 original size) and hardly affect performance.

5 Conclusion

In this paper we present SVCCA, a general method which allows for comparison of the learned distributed representations between different neural network layers and architectures. Using SVCCA we obtain novel insights into the learning dynamics and learned representations of common neural network architectures. These insights motivated a new Freeze Training technique which can reduce the number of flops required to train networks and potentially even increase generalization performance. We observe that CCA similarity can be a helpful tool for interpretability, with sensitivity to different classes reflecting their semantic properties. This technique also motivates a new algorithm for model compression. Finally, the "lower layers learn first" behavior was also observed for recurrent neural networks as shown in Figure App.6 in the Appendix.

References

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.

[2] David Eigen, Jason Rolfe, Rob Fergus, and Yann LeCun. Understanding deep architectures using a recursive convolutional network. *arXiv preprint arXiv:1312.1847*, 2013.

[3] Manaal Faruqui and Chris Dyer. Improving vector space word representations using multilingual correlation. Association for Computational Linguistics, 2014.

[4] Nathan Halko, Martinsson Per-Gunnar, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Rev.*, 53:217–288, 2011.

[5] D. R. Hardoon, S. Szedmak, and J. Shawe-Taylor. Canonical correlation analysis: An overview with application to learning methods. *Neural Computation*, 16:2639–2664, 2004.

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. *CoRR*, abs/1512.03385, 2015.

[7] Geoffrey Hinton, Li Deng, Dong Yu, George E. Dahl, Abdel-rahman Mohamed, Navdeep Jaitly, Andrew Senior, Vincent Vanhoucke, Patrick Nguyen, Tara N Sainath, et al. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal Processing Magazine*, 29(6):82–97, 2012.

[8] Roger A Horn and Charles R Johnson. *Matrix analysis*. Cambridge university press, 1985.

[9] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In *Advances in neural information processing systems*, pages 1097–1105, 2012.

[10] Karel Lenc and Andrea Vedaldi. Understanding image representations by measuring their equivariance and equivalence. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 991–999, 2015.

[11] Y. Li, J. Yosinski, J. Clune, H. Lipson, and J. Hopcroft. Convergent Learning: Do different neural networks learn the same representations? In *International Conference on Learning Representations (ICLR)*, May 2016.

[12] Yixuan Li, Jason Yosinski, Jeff Clune, Hod Lipson, and John Hopcroft. Convergent learning: Do different neural networks learn the same representations? In *Feature Extraction: Modern Questions and Challenges*, pages 196–212, 2015.

[13] Aravindh Mahendran and Andrea Vedaldi. Understanding deep image representations by inverting them. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5188–5196, 2015.

[14] Grégoire Montavon, Mikio L Braun, and Klaus-Robert Müller. Kernel analysis of deep networks. *Journal of Machine Learning Research*, 12(Sep):2563–2581, 2011.

[15] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps. *arXiv preprint arXiv:1312.6034*, 2013.

[16] David Sussillo, Mark M Churchland, Matthew T Kaufman, and Krishna V Shenoy. A neural network that finds a naturalistic solution for the production of muscle activity. *Nature neuroscience*, 18(7):1025–1033, 2015.

[17] Christian Szegedy, Wojciech Zaremba, Ilya Sutskever, Joan Bruna, Dumitru Erhan, Ian Goodfellow, and Rob Fergus. Intriguing properties of neural networks. *arXiv preprint arXiv:1312.6199*, 2013.

[18] Yonghui Wu, Mike Schuster, Zhifeng Chen, Quoc V. Le, Mohammad Norouzi, Wolfgang Macherey, Maxim Krikun, Yuan Cao, Qin Gao, Klaus Macherey, et al. Google's neural machine translation system: Bridging the gap between human and machine translation. *arXiv preprint arXiv:1609.08144*, 2016.

[19] Jason Yosinski, Jeff Clune, Anh Nguyen, Thomas Fuchs, and Hod Lipson. Understanding neural networks through deep visualization. In *Deep Learning Workshop, International Conference on Machine Learning (ICML)*, 2015.